Saboot

Satellite Evidence for Environmental Complaints

Event: Environmental Hacks, Bharat Builds Tour (WeMakeDevs and AWS), October 8 to 11, 2026
Track: Waste and Energy
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Problem

Residents who report illegal dumping or waste burning usually have one photo, and complaints are closed without action. New national Solid Waste Management Rules took effect on April 1, 2026, but few people know what they prohibit or whom to hold responsible. Years of satellite history exist for every location in India, yet ordinary citizens cannot use it as evidence.

Solution

Saboot ("proof") turns a map pin into a filing-ready complaint backed by satellite evidence and the rule that applies.

1. **Pin.** The resident marks the spot and can add a photo.
2. **History.** The app shows dated satellite images of that spot and any fires detected there.
3. **Lookalikes.** AI finds places across the district that look the same from space and shows them as a hotspot map.
4. **Draft.** An AI agent retrieves the applicable rule and the responsible authority, then writes a one-page complaint in Hindi or English with the evidence attached. The resident reviews it and files it.

Technology used

Layer	Technology
Web app	React and MapLibre GL, hosted on AWS Amplify
API	Amazon API Gateway and AWS Lambda (Python)
Satellite imagery	Sentinel-2 open data read directly from Amazon S3
Fire detections	NASA FIRMS active fire data
Lookalike search	Earth Genome open embeddings (384 values per 320 m patch), searched with OpenSearch vector search
Rule retrieval	OpenSearch hybrid search (BM25 plus vectors) over the 2026 rules and tribunal orders
Batch processing	Apache Spark (PySpark) on Amazon EMR Serverless filters and indexes the embedding files. The same job scales from one district to all of India
Data lake	Site time series stored as Parquet on Amazon S3, partitioned by district and month, queried with Amazon Athena
AI agent	Strands Agents SDK with an LLM on Amazon Bedrock
App storage	Amazon DynamoDB for cases, Amazon S3 for thumbnails and PDFs

Big data concepts: distributed batch processing with Spark, and partitioned columnar storage in a data lake.

What is new

Complaint apps and satellite search tools both exist. Saboot joins them into one flow: pin, satellite evidence, similar sites and a cited complaint. Every rule in the draft comes from a retrieved document shown beside it, so the text reflects the 2026 rules and not an AI model’s memory of the old ones.

Scope for the event

- **Must have:** pin, satellite history, fires, rule retrieval and complaint draft, for Jodhpur district.
- **Should have:** lookalike search with a hotspot map.
- **If time allows:** Hindi voice input and a public case tracker.

Known limits

- Each embedding covers a 320 m patch, so only large dump and burn sites can be found. Roadside piles cannot.
- Satellite images show that a site exists and when it changed, not who is responsible. Drafts describe a location and never name a person.
- The draft is not legal advice. The resident files it by hand, and nothing is submitted automatically.